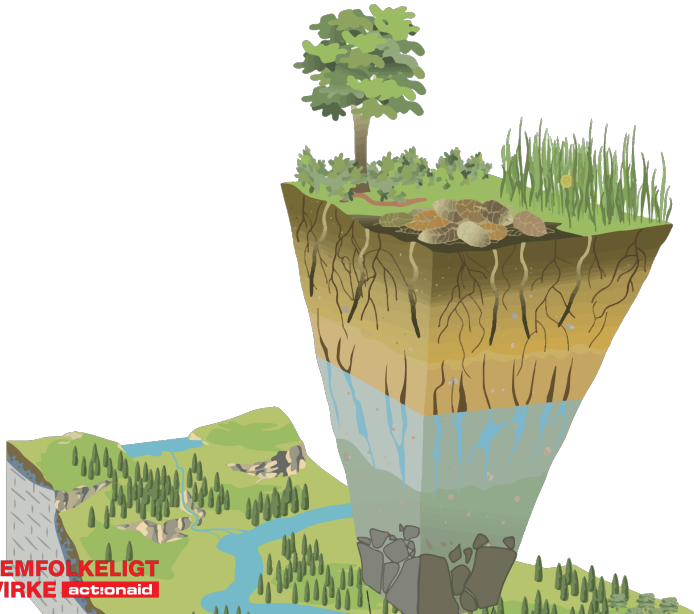


# Don't Shit where You Drill!



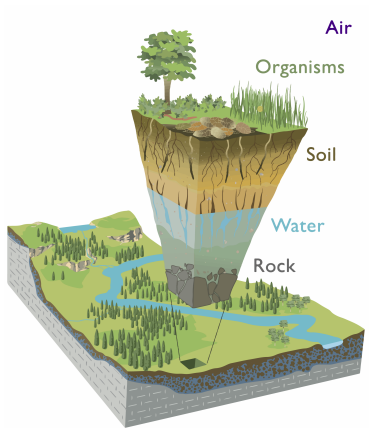
# Don't Shit where You Drill!

Michel Hardenberg

PhD student at AU - Quantifying the Vadose Zone  
Department of Electrical and Computer Engineering

**MELLEMFOLKELIGT  
SAMVIRKE** **act:onaid**

CLIMATE JUSTICE SOCIETY  
JUNE 11, 2026



*Illustration of soil layers in the near surface [1].*

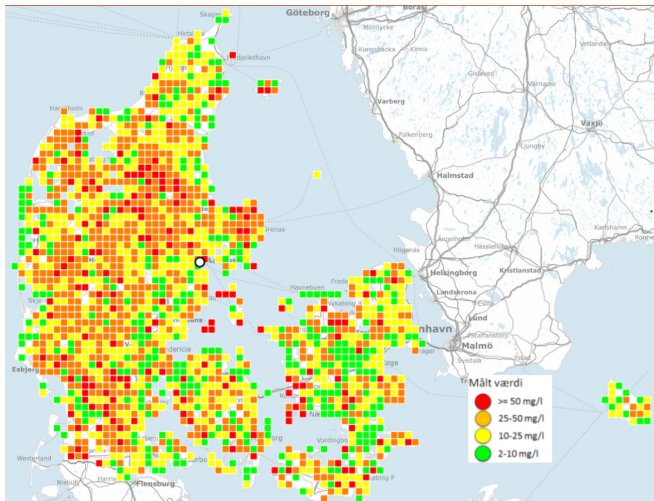
# ToC

## Part One:

- ▶ What are the culprits?
- ▶ How are we looking?
- ▶ What is the government doing? Promising?

# Nitrate - the Classic

# NITRATE MEASUREMENTS



Nitrate pollution in Denmark

(<https://www.geus.dk/produkter-ydelser-og-faciliteter/data-og-kort/grundvandskort-og-data>)

*“Nitrate in drinking water is a contaminant which can affect human health and has been associated with an increased risk of, amongst other diseases, colorectal cancer. [...] The analysis shows that a lower limit of 9.25 mg/L would decrease the annual number of colorectal cancer cases by 72 [...] and by an additional 55 % [...] for a stricter limit of 3.87 mg/L.”*

# PFAS - the Immortal

# WHERE DO THEY LIVE?

**Where DON'T they live!?** PFAS are found in water, air, fish, and soil..

# WHERE DO THEY LIVE?

**Where DON'T they live!?** PFAS are found in water, air, fish, and soil..Also, in blood of people, animals, in food products<sup>1</sup>

# ToC

## Part Two:

- ▶ Case studies in the global south
- ▶ How are we as Europeans involved?
- ▶ How is climate change changing the picture?

# REFERENCES I



J. Chorover, R. Kretzschmar, F. Garcia-Pichel, and D. L. Sparks, “Soil Biogeochemical Processes within the Critical Zone,” *Elements*, vol. 3, pp. 321–326, Oct. 2007.